Suppose X is a discrete random variable taking the values $x_1, x_2, x_3, \ldots$ such that $x_1 < x_2, < x_3 \cdots$ and $F(x_i)$ is the distribution function. Then the probability mass function $f(x_i)$ is given by

$$f(x_i) = F(x_i) - F(x_{i-1}), \quad i = 1, 2, 3, \ldots$$

Note

The jump of a function $F(x)$ at $x = a$ is $\left|F(a^+) - F(a^-)\right|$. Since F is non-decreasing and continuous to the right, the jump of a cumulative distribution function F is $P(X = x) = F(x) - F(x^-)$.

Here the jump (because of discontinuity) acts as a probability. That is, the set of discontinuities of a cumulative distribution function is at most countable!

Example 11.9

Find the probability mass function $f(x)$ of the discrete random variable X whose cumulative distribution function $F(x)$ is given by

$$F(x) = \begin{cases} 0 & -\infty < x < -2 \\ 0.25 & -2 \le x < -1 \\ 0.60 & -1 \le x < 0 \\ 0.90 & 0 \le x < 1 \\ 1 & 1 \le x < \infty \end{cases}$$

Also find (i) $P(X < 0)$ and (ii) $P(X \ge -1)$.

Solution

Since X is a discrete random variable, from the given data, X takes on the values

$-2, -1, 0,$ and 1.

For discrete random variable X, by definition, we have $f(x) = P(X = x)$

Therefore left hand limit of $F(x)$ at $x = -2$ is $F(-2^-)$

$$f(-2) = P(X = -2) = F(-2) - F(-2^-) = 0.25 - 0 = 0.25.$$

Similarly for other jump points, we have

$$f(-1) = P(X = -1) = F(-1) - F(-2) = 0.60 - 0.25 = 0.35.$$

$$f(0) = P(X = 0) = F(0) - F(-1) = 0.90 - 0.60 = 0.30,$$

$$f(1) = P(X = 1) = F(1) - F(0) = 1 - 0.90 = 0.10.$$

Therefore the probability mass function is

x	-2	-1	0	1
$f(x)$	0.25	0.35	0.30	0.10

The distribution function $F(x)$ has jumps at $x = -2, -1, 0,$ and 1. The jumps are respectively $0.25, 0.35, 0.30,$ and 0.1 is shown in the figure given below.

These jumps determine the probability mass function

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